CHAPTER 01 :

INTRODUCTION

1.1 INTRODUCTION:

Autonomous navigation for mobile robotics and Unmanned Aerial Vehicles (UAVs) relies on an ability to navigate unmapped spaces without human intervention. While path planners map optimal trajectories using pre-computed environmental data, they face high latency and memory constraints when trying to adapt to sudden changes. This limitation drives the development of reactive obstacle avoidance, which discards heavy 3D mapping and machine learning architectures in favor of directly converting immediate sensory data into real-time velocity commands. By eliminating high computation, reactive systems prioritize minimal processing latency, making them ideal for low-altitude operations and compact aerial platforms.

However, relying on a single sensory modality presents a single point of failure because different sensors fail on different materials. Ultrasonic sensors calculate distance using high frequency sound waves. They are highly effective against solid, flat surfaces like concrete, but fail significantly on sound-absorbing materials like porous fabrics, leading to critical overestimation errors where the drone detects an obstacle as farther away than it actually is. Conversely, Time-of-Flight (ToF) infrared sensors measure photon travel times to provide high-speed spatial resolution. Yet, they suffer from data loss or severe underestimation when encountering optically non-cooperative surfaces, such as transparent glass or specularly reflective metals that disperse the infrared beam.

To overcome these individual vulnerabilities, this project develops a multi-sensor data fusion system running on a low-power, dual-core ESP32 microcontroller. By synthesizing data streams from dual Ultrasonic sensors and a Time of flight sensor, the architecture establishes a robust cross-checking framework. Kalman filters are used running on the embedded firmware to suppress noise from the sensors.

The results are then combined using a calibrated weighted-average model. This model applies optimized base weights that favor the underestimating nature of the ToF sensor, which naturally causes the vehicle to stop safely early rather than collide. This approach provides highly dependable obstacle avoidance on an affordable, low-power microcontroller without the need for extensive training data or intense graphics processing power.

1.2 OBJECTIVE:

The primary objective of this research is the designing, evaluation, and implementation of a low-latency, embedded reactive obstacle avoidance subsystem. By utilizing intelligent multi-sensor data fusion, the system aims to ensure robust vehicle safety across diverse material interfaces. The specific technical objectives are outlined as follows:

- Development of a compact embedded architecture based on the ESP32 microcontroller to interface with the multi-voltage sensor grid. This involves the integration of voltage divider networks to scale 5V Echo signals to the 3.3V logic thresholds of the ESP32, preventing hardware degradation.
- Execution of systematic experimental trials to characterize the distance telemetry of Ultrasonic and TOF sensors against six environmental materials (wood, aluminum, glass, concrete, plastic, and fabric) at predefined calibration intervals of 25 cm, 50 cm, and 75 cm.
- Derivation of linear regression-based calibration curves to correct systematic sensor drift. By analyzing raw data distributions against physical benchmarks, offsets and scale factors are extracted for real-time firmware compensation.
- Implementation of Kalman filtering algorithms within the ESP32 firmware loop to eliminate the high-frequency fluctuations caused by electronic noise and structural vibrations.
- Development of a deterministic fusion layer employing a calibrated Tof - Ultrasonic weighting model. A safety override mechanism is integrated to trigger when sensor disagreement exceeds 3.0 cm, prioritizing the most conservative reported hazard.
- Deployment of the avoidance pipeline within the ESP32's computational limits. This architecture ensures real-time velocity modulation without the requirement of 3D mapping or high-power machine learning hardware.

CHAPTER 02 :

LITERATURE REVIEW

Evaluating different approaches to state estimation and distance tracking is essential for building a reliable, low-latency obstacle avoidance system.These are the methodologies in state estimation, sensor characterization, and multi-modal sensor fusion that apply to autonomous platforms.

2.1 Overview of Existing Technologies

2.1.1 Kalman Filtering in Altitude Estimation

Precise distance and altitude estimation in autonomous aerial vehicles is fundamentally limited by the noise characteristics of low-cost sensors. Raw sensor readings from devices such as barometers, IMUs, and contain noise and systematic bias that, if passed directly to a control loop, produce unstable or oscillatory behaviour. A well-established solution to this problem is the Kalman filter, a recursive, efficient estimator that models both the uncertainty of the system state to produce an optimal fused estimate.

The key insight applicable to the present work is that statistical filtering does not require high computational resources to be effective, a well-parameterized 1D or 2D Kalman filter applied independently to each sensor before fusion is sufficient to suppress noise while preserving the dynamic response needed for real-time obstacle avoidance

2.1.2 Distance Measurement Sensors and Multi-Sensor Fusion

The external environment perceived by an autonomous platform is only as reliable as the sensors used to measure it. A comparative analysis of mainstream range-finding technologies including laser, infrared Time-of-Flight, Ultrasonic, Millimetre-wave radar, and Vision-based systems reveals that no single sensing modality performs uniformly well across all operating conditions.

The conclusion drawn across multiple studies is consistent: a single sensor modality, while adequate under controlled conditions, exhibits predictable failure modes in real-world unstructured environments. This motivates a multi-sensor fusion approach in which the complementary strengths of each modality are exploited specifically, pairing an ultrasonic

sensor with a Time-of-Flight sensor to provide mutual redundancy across a broader range of surface materials.

2.1.3 Accuracy Enhancement in Low-Cost Microcontrollers

Deploying sensor fusion algorithms on low-cost microcontrollers introduces additional constraints beyond algorithmic design. Budget-friendly sensors that interface with platforms such as Arduino or ESP32 typically exhibit greater manufacturing variance, temperature sensitivity, and non-linear response curves compared to industrial-grade alternatives.

Empirical characterization of sensor behaviour across multiple known distances allows the construction of a correction model that accounts for both fixed bias and distance-proportional error. This characterization step also yields the measurement noise variance R, a critical parameter for Kalman filter tuning. Research in this area confirms that even modest calibration procedures applied to commodity sensors can bring their accuracy to within acceptable bounds for closed-loop control, provided the residual noise is subsequently managed by an appropriate filtering stage.

2.2 Summary of Literature Gaps and Project Focus

Existing literature establishes the theoretical foundations of Kalman filtering and multi-sensor fusion but leaves practical gaps in material-specific sensor characterization, safety-oriented weight assignment, and single-sensor failure handling on embedded platforms. This project addresses these gaps by empirically quantifying how six real-world surface materials affect the bias and accuracy of both ultrasonic and Time-of-Flight sensors, using those findings to assign fusion weights that prioritize collision-avoiding conservative behaviour, and implementing a deterministic safety override rule that handles material-induced sensor failures all within a complete, validated pipeline running on a low-cost ESP32 microcontroller.

Author(s)	Title	Publication details	Description
Seyedmilad Komarizadehasl, Behnam Mobaraki, Haiying Ma,	Low-Cost Sensors Accuracy Study and Enhancement	Received: 24 February 2022 Accepted: 18 March 2022 Published: 21	Budget-friendly sensors provide the benefit of straightforward integration with affordable microcontrollers like the Arduino platform. By utilizing Arduino

Jose-Antonio Lozano-Galant and Jose Turmo	Strategy	March 2022 Appl. Sci. 2022, 12(6), 3186; https://doi.org/10.3390/app12063186	technology, a dependable and low-cost acquisition framework can significantly decrease expenses associated with data collection and surveillance, facilitating the feasibility of extended monitoring periods. This research presents a wireless, internet-enabled data acquisition architecture that utilizes a Raspberry Pi in conjunction with multiple Arduino units acting as signal conditioners. Furthermore, this work examines how clustering comparable sensor types can enhance the collective precision of a sensor array characterized by undefined accuracy parameters.
Jinbo Li	Study of the Principles and Applications of Common Distance Measuring Sensors	Greene's College Oxford, Oxford, The United Kingdom Proceedings of CONF-FMCE 2025 Symposium: Semantic Communication for Media Compression and Transmission DOI: 10.54254/2755-2721/2025.GL2493	Distance measurement sensors play an important role in unmanned driving, robotics, industrial control, and other fields. This article discusses the classification and fusion of distance measurement sensors and introduces the working principles and characteristics of various mainstream sensors, such as laser, infrared, ultrasonic, millimeter wave radar, and camera, respectively. This article introduces the differences in distance measurement accuracy, response speed, environmental adaptability, and focuses on the combination solutions to achieve fusion and the advantages of sensor fusion technology. The research results show that a single sensor works well under specific conditions, but is easily affected in complex environments, while multi-sensor fusion can effectively improve the overall ability of the system.
Muhammad Asad Bashir, Fahad	Kalman Filter Based Sensor Fusion for	Department of Electrical Engineering,	Stabilization of Aerial Vehicles requires altitude estimation. In this paper sensor fusion method is

Mumtaz Malik, Zeeshan Ali Akbar and Muhammad Uzair	Altitude Estimation of Aerial Vehicle	National University of Sciences & Technology (NUST) College Of Electrical & Mechanical Engineering, Rawalpindi ,Pakistan IOP Conf. Series: Materials Science and Engineering (2020) doi:10.1088/1757-899X/853/1/012034	developed for aerial vehicles which focuses on estimation of the altitude. Altitude measurement using sensors is too noisy and biased, so Kalman filter based sensor fusion is used to estimate altitude. The proposed method is divided into two parts: attitude estimation and altitude estimation. The method is implemented using IMU and barometer sensors. The results show that altitude is estimated accurately and errors are modelled in few seconds after initialization.
Angelo Maria Sabatini and Vincenzo Genovese	A Sensor Fusion Method for Tracking Vertical Velocity and Height Based on Inertial and Barometric Altimeter Measurements	The BioRobotics Institute, Scuola Superiore Sant'Anna, Viale Rinaldo Piaggio, Pontedera 34 56025, Pisa, Italy Received: 26 January 2014; in revised form: 8 July 2014 / Accepted: 21 July 2014 / Published: 24 July 2014	A sensor fusion method was developed for vertical channel stabilization by fusing inertial measurements from an Inertial Measurement Unit (IMU) and pressure altitude measurements from a barometric altimeter integrated in the same device (baro-IMU). An Extended Kalman Filter (EKF) estimated the quaternion from the sensor frame to the navigation frame; the sensed specific force was rotated into the navigation frame and compensated for gravity, yielding the vertical linear acceleration; finally, a complementary filter driven by the vertical linear acceleration and the measured pressure altitude produced estimates of height and vertical velocity.

Table 2.1 Literature Survey Details

CHAPTER 03 :

BLOCK DIAGRAM

To determine real-time performance of the sensors, the system is divided into two separate operational pipelines running concurrently on the ESP32 microcontroller. This dual-subsystem design separates internal vehicle kinematics from external telemetry, integrating them into a unified reactive control loop.

The system's architectural layout is illustrated below.

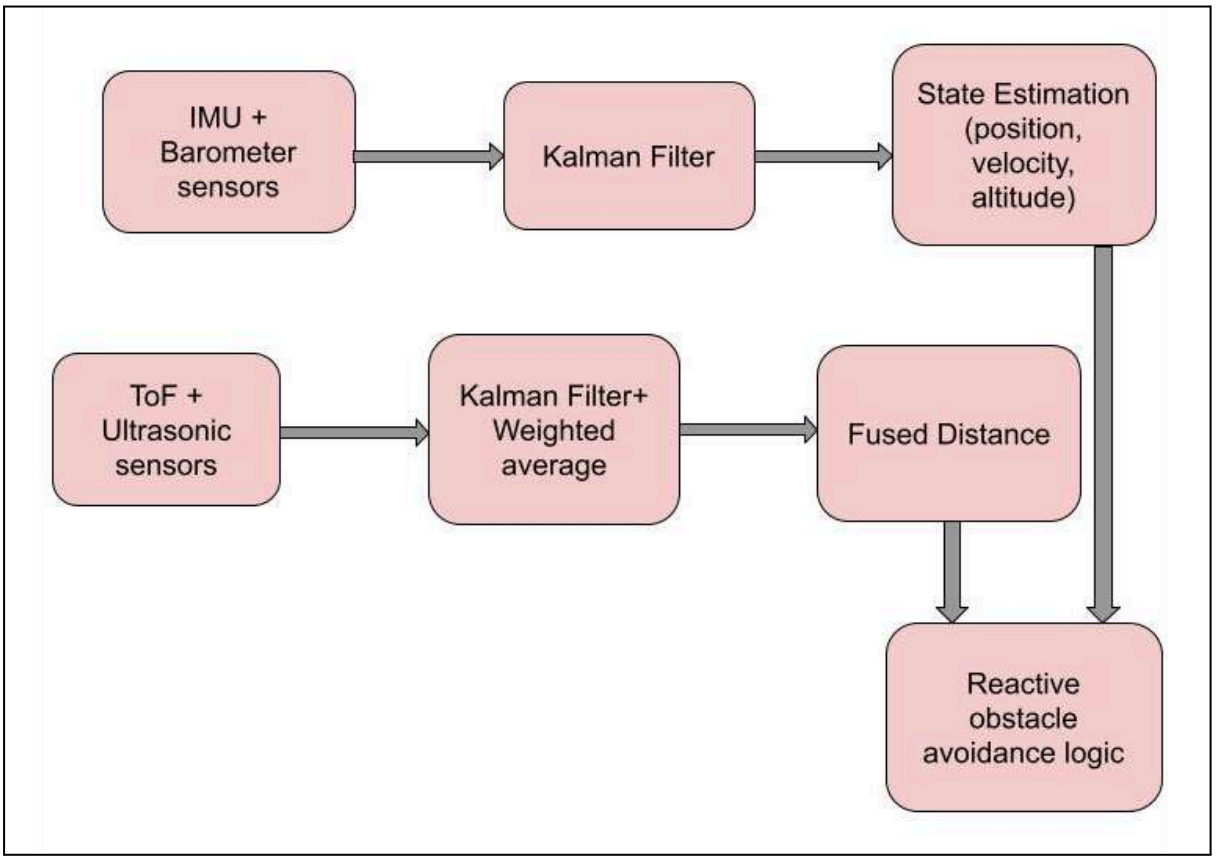


Fig 3.1 Block diagram of Reactive Obstacle Avoidance

3.1 State Estimation Subsystem:

The first pipeline focuses on tracking the physical state and spatial coordinates of the autonomous platform. It processes raw inputs from an MPU9250 Inertial Measurement Unit and a barometric pressure altimeter to generate stable, drift-free positional feedback.

- IMU + Barometer Sensors: The IMU tracks high-frequency angular velocities and linear accelerations across three axes, while the barometer measures atmospheric pressure variations to provide an absolute altitude reference.
- Kalman Filter: Raw data from these sensors is prone to high-frequency noise, and localized atmospheric drift. A Kalman filter sequentially models these errors. It fuses high-frequency accelerometer data with low-frequency barometric references to eliminate signal lag and integration drift.
- State Estimation Output: This provides a stabilized, real-time 9DoF pose, vertical velocity, and altitude estimate.

3.2 Obstacle Avoidance Subsystem

The secondary pipeline functions independently, mapping external obstructions to calculate a precise, noise-filtered proximity vector in real time.

- ToF + Ultrasonic Sensors: Proximity data is gathered simultaneously using a VL53L0X Time-of-Flight sensor and dual HC-SR04 ultrasonic sensors. The ToF sensor measures distance by emitting a pulse of light and calculating how long it takes for that light to bounce off an object and return to the sensor to provide highly localized spatial resolution, while the ultrasonic array measures sound wave reflections over a broader field of view.
- Kalman Filter + Weighted Average: The raw distance measurements are first processed through individual, parallel Kalman filters to smooth out signal spikes and sensor inconsistencies. Once filtered, the data streams are combined using a static weighted-average model.
- Fused Distance Output: This block constantly outputs a single, consolidated distance value. If a significant disagreement occurs between the optical and acoustic channels—such as when encountering sound-absorbing fabric or transparent glass—the fusion layer automatically bypasses the base weights and outputs the absolute shorter reported distance to ensure maximum safety.

3.3 Reactive Obstacle Avoidance Logic

The final block acts as the central decision-making hub. By taking the real-time velocity/position vectors from the State Estimation subsystem and matching them against the

Fused Distance vector from the Obstacle Avoidance subsystem, the controller dynamically calculates changes in trajectory.

Because this block bypasses path-planning, 3D mapping, or machine learning inferences, it directly modulates the vehicle's velocity commands with minimal processing latency. This ensures the autonomous platform reacts instantly to local obstructions, slowing down or changing course to maintain a protective boundary in cluttered environments.

CHAPTER 04 :

CIRCUIT DIAGRAM

The complete physical layout and wiring connections for the project are mapped out in the schematic below. This diagram illustrates exactly how the power tracks, digital timing lines, and I2C communication buses link the sensors to the central ESP32.

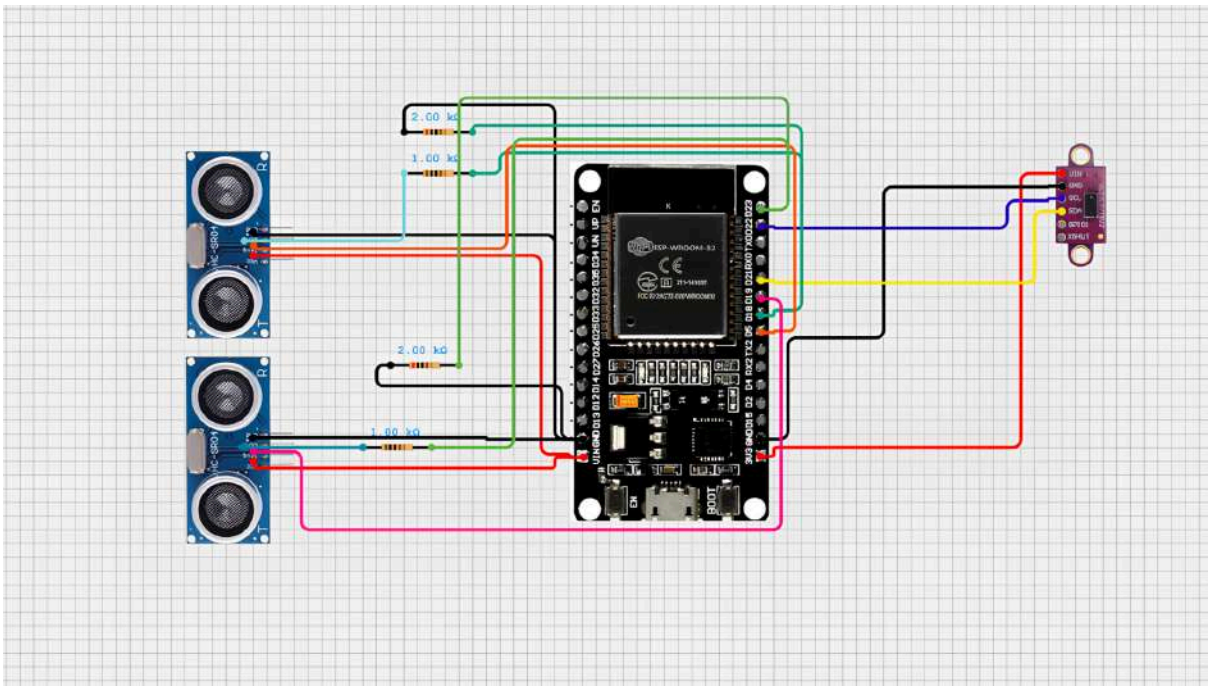


Fig 4.1 Circuit diagram of Sensor Fusion

To translate the system's logic into functional hardware, the circuit must maintain stable power delivery while handling clean signal transmission across mismatched logic levels. The setup centers on an ESP32 development board interfacing with dual HC-SR04 ultrasonic sensors and a single VL53L0X Time-of-Flight (ToF) sensor.

4.1 Power Domain Allocation

The hardware operates across two distinct voltage rails: a 5V domain for the acoustic transducers and a 3.3V domain for the microcontroller core and I2Cdata bus.

- 5V Sensor Power: The VCC pins of both HC-SR04 ultrasonic sensors connect directly to the VIN pin of the ESP32. When powered via USB, the VIN pin acts as a

direct 5V pass-through from the cable. This full 5V supply is necessary because running these sensors at 3.3V weakens the acoustic pings, leading to missed echoes and unstable readings.

- 3.3V Sensor Power: The VL53L0X ToF sensor runs natively on 3.3V and draws its power directly from the ESP32's onboard 3V3 regulator pin.
- Common Ground: All sensor ground lines route back to the ESP32's GND pins to establish a unified reference point and prevent floating signal errors.

4.2 Signal Conditioning via Voltage Dividers

While the ESP32 can safely trigger the ultrasonic sensors using its 3.3V output pins (GPIO 12 and 14), it cannot safely handle the returning signals. The HC-SR04 outputs a 5V pulse on its Echo pin, which will degrade or instantly fry the ESP32's 3.3V tolerant GPIO pins if connected directly.

To safely drop this voltage, a passive resistor divider network is wired inline with each Echo pin. By placing a 1k ohm resistor in series with a 2k ohm tied to ground, the voltage seen by the ESP32 input pins (GPIO 13 and 27) is stepped down:

$$Vout = Vin\left(\frac{R2}{R1+R2}\right)$$

This 3.33V signal is safely within the microcontroller's operating limits, ensuring reliable pulse-width timing without risking hardware damage.

4.3 Communication Bus

The VL53L0X ToF sensor communicates via the I2C protocol rather than raw digital pulses. It connects directly to the ESP32's dedicated hardware I2Cpins:

- SDA (Serial Data Line): Wired to GPIO 21 for bidirectional data transfer.
- SCL (Serial Clock Line): Wired to GPIO 22 to synchronize data frames.

By using digital pins for the ultrasonic sensors and the I2C bus for the ToF sensor, the ESP32 can read all three sensors at the same time without slowing down.

CHAPTER 05:

WORKING OF THE PROJECT

5.1. Hardware Setup and Interfacing

5.1.1 Platform

The ESP32 development module was selected as the central processing unit for this project due to its dual-core 240 MHz processor, native I2C and GPIO support, and 3.3 V logic levels compatible with both sensor types. Serial communication was established at 115200 baud for real-time data logging and monitoring.

5.1.2 HC-SR04 Ultrasonic Sensors (×2)

Two HC-SR04 ultrasonic sensors were interfaced with the ESP32. The HC-SR04 operates at 5V but the ESP32's GPIO pins are 3.3 V tolerant only. A voltage divider was therefore placed on each ECHO line to step the 5 V return signal down to a safe 3.3 V level.

Connection	Sensor 1	Sensor 2
VCC	ESP32 5V/VIN	ESP32 5V/VIN
GND	ESP32 GND	ESP32 GND
TRIG	GPIO 5	GPIO 23
ECHO	GPIO 18 (via 1kΩ/2kΩ divider)	GPIO 19 (via 1kΩ/2kΩ divider)

Table 5.1.2 Pin details for Ultrasonic sensor

The voltage divider uses a 1 kΩ resistor between the ECHO pin and the ESP32 GPIO, and akΩ resistor between the ESP32 GPIO and GND. This produces a divided voltage of 5 V × (2/3) = 3.33 V at the GPIO input, which is within the safe operating range of the ESP32.

5.1.3 Adafruit VL53L0X Time-of-Flight Sensor

The VL53L0X was interfaced via the I2C bus. The Adafruit breakout board includes an onboard voltage regulator, making it compatible with both 3.3 V and 5 V supply rails. No additional level shifting was required.

VL53L0X Pin	ESP32 Pin
VIN	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22
XSHUT	Not connected (always on)

Table 5.1.3 Pin details for Time of Flight sensor

The sensor was configured in default mode for short-to-medium range operation (up to approximately 120 cm), which offers better accuracy at the test distances of 25 cm, 50 cm, and 75 cm compared to long-range mode.

5.2. Sensor Calibration

Before any filtering or fusion was applied, each sensor was individually calibrated to correct for systematic measurement errors. Calibration was performed by collecting 50 samples at each of three known distances and computing a scale factor and offset to bring the average reading into agreement with the true distance.

5.2.1 Calibration Methodology

For each sensor and each known distance d, 50 samples were collected and averaged. The calibration parameters were then computed as:

$$\textbf{Scale Factor} = \textbf{Known Distance} \;/\; \textbf{Average Raw Reading}$$

$$\textbf{Offset} = \textbf{Known Distance} - \textbf{Average Raw Reading}$$

A scale factor greater than 1 indicates the sensor under-reads and the correction stretches the reading upward. A scale factor less than 1 indicates the sensor over-reads. The offset corrects any fixed additive bias.

5.2.2 Temperature Compensation for Ultrasonic Sensors

The speed of sound in air varies with temperature. Rather than hardcoding a fixed value, the speed of sound was computed from the ambient temperature using:

$$v = (331.3 + 0.606 \times T) / 10000 \quad cm/\mu s$$

where T is the ambient temperature in degrees Celsius. At the test environment temperature of 28°C, this gives v = 348.2 m/s, compared to the commonly hardcoded value of 343 m/s at 20°C — a difference of approximately 1.5%, which at 100 cm translates to a 1.5 cm error if uncorrected.

5.2.3 Calibration Results

5.2.3.1 HC-SR04 Ultrasonic (both sensors):

Known Distance	Avg Raw (cm)	Offset (cm)	Scale Factor
25 cm	24.43	+0.57	1.023
50 cm	48.87	+1.13	1.023
75 cm	73.30	+1.70	1.023

Table 5.2.3.1 Calibration results of Ultrasonic (both sensors)

The ultrasonic sensors showed a consistent scale factor of 1.023 across all distances, indicating a proportional under-read of approximately 2.3%. The final calibration parameters applied were: Scale = 1.023, Offset = 0.0.

5.2.3.2 VL53L0X Time-of-Flight:

Known Distance	Avg Raw (cm)	Offset (cm)	Scale Factor
25 cm	27.08	−2.08	0.9233
50 cm	52.60	−2.60	0.9506
75 cm	76.84	−1.84	0.9761

Table 5.2.3.2 Calibration results of Time of Flight sensor

The ToF sensor exhibited a consistent negative offset — it consistently over-reads the raw distance by approximately 2 cm across all test distances. Since the scale factor was inconsistent across distances (varying from 0.92 to 0.98), an offset-only correction was applied: Scale = 1.0, Offset = −2.17 cm (average of the three offset values). This approach avoids over-correcting at longer distances where the scale factor was already close to 1.

5.3 Kalman Filtering

Following calibration, a 1D Kalman filter was applied independently to each sensor to suppress residual measurement noise. The Kalman filter provides an optimal estimate of the true distance by combining the incoming noisy measurement with a predicted state, weighted by the relative uncertainty of each.

5.3.1 Filter Equations

The 1D Kalman filter implemented for each sensor follows the standard predict-update cycle:

Step	Equation	Description
Prediction	$P = P + Q$	Grow uncertainty by process noise Q
Kalman Gain	$K = P / (P + R)$	Balance prediction vs measurement trust
Update	$x = x + K \times (z - x)$	Blend prediction with new measurement z
Covariance	$P = (1 - K) \times P$	Reduce uncertainty after update

Table 5.3.1 Kalman Filter Equations

where **Q** is the process noise (how much the true distance changes between readings), **R** is the measurement noise (the sensor's variance), **x** is the current state estimate, and **P** is the estimation error covariance. The parameters used were $Q = 0.05$ and $R = 0.3$, tuned empirically to balance noise suppression with responsiveness to real distance changes.

5.3.2 Why Kalman Before Fusion

Each sensor is Kalman filtered independently before any fusion is performed. This ensures that the fusion stage receives clean, noise-characterised inputs rather than combining two noisy raw signals, which would compound rather than reduce uncertainty. The per-sensor filtering also yields the measurement noise variance R for each sensor independently, which is used to inform the fusion weight assignment.

5.4 Experimental Results — Material Testing

Both sensors were tested against six surface materials at three known distances: 25 cm, 50 cm, and 75 cm. For each combination, the raw reading, calibrated reading, and Kalman-filtered reading were recorded. The results are presented in the tables below.

5.4.1 Results at 25 cm

Material	Raw (cm)	Calibrated (cm)	Kalman (cm)	Error (cm)
Wood	26.10	23.82	23.52	-1.48
Metal	24.50	22.21	21.95	-3.05
Glass	24.40	22.11	21.85	-3.15
Concrete/Wall	27.10	24.83	24.90	-0.10
Coloured Plastic	25.80	23.52	23.30	-1.70
Fabric	26.20	23.92	23.96	-1.04

Table 5.4.1a —Results of ToF sensor at 25 cm (Known: 25.0 cm)

Material	Raw U1	KF U1	Raw U2	KF U2	Fused KF	Error (cm)
Wood	24.3	25.1	24.0	25.3	25.2	0.20
Metal	24.3	25.3	23.7	25.1	25.2	0.20
Glass	24.3	25.3	24.3	25.7	25.5	0.50
Concrete/Wall	23.7	24.9	23.7	25.2	25.0	0.00
Coloured Plastic	24.3	25.3	24.0	25.6	25.4	0.40
Fabric	26.7	26.0	26.6	26.4	26.2	1.20

Table 5.4.1b —Results of Ultrasonic sensors at 25 cm (Known: 25.0 cm)

5.4.2 Results at 50 cm

Material	Raw (cm)	Calibrated (cm)	Kalman (cm)	Error (cm)
Wood	52.20	50.05	48.90	-1.10
Metal	49.60	47.44	47.40	-2.60
Glass	47.20	46.03	46.50	-3.50
Concrete/Wall	52.20	50.15	50.20	0.20
Coloured Plastic	52.20	50.05	48.90	-1.10
Fabric	51.90	50.60	49.60	-0.40

Table 5.4.2a —Results of ToF sensor at 50 cm (Known: 50.0 cm)

Material	Raw U1	KF U1	Raw U2	KF U2	Fused KF	Error (cm)
Wood	48.4	50.2	48.6	50.2	50.5	0.50
Metal	48.2	50.0	48.4	50.3	50.2	0.20
Glass	48.2	50.0	48.4	51.5	50.7	0.70
Concrete/Wall	48.5	50.2	48.2	50.3	50.3	0.30
Coloured Plastic	47.6	50.4	48.6	50.7	50.6	0.60
Fabric	49.3	51.4	50.3	52.8	52.1	2.10

Table 5.4.2b — Results of Ultrasonic sensors at 50 cm (Known: 50.0 cm)

5.4.3 Results at 75 cm

Material	Raw (cm)	Calibrated (cm)	Kalman (cm)	Error (cm)
Wood	76.20	74.17	74.68	-0.32
Metal	76.50	74.70	73.85	-1.15
Glass	72.20	70.15	70.29	-4.71
Concrete/Wall	76.40	74.87	75.09	0.09
Coloured Plastic	72.60	71.16	73.40	-1.60
Fabric	72.70	75.68	75.06	0.06

Table 5.4.3a —Results of ToF sensor at 75 cm (Known: 75.0 cm)

Material	Raw U1	KF U1	Raw U2	KF U2	Fused KF	Error (cm)
Wood	72.6	75.1	72.7	75.4	75.3	0.30
Metal	72.7	75.2	73.1	75.9	75.4	0.40
Glass	72.5	74.9	71.5	74.6	74.8	-0.20
Concrete/Wall	72.7	75.1	72.8	75.5	75.3	0.30
Coloured Plastic	73.1	75.6	73.1	75.9	75.7	0.70
Fabric	73.0	75.2	74.4	77.1	76.2	1.20

Table 5.4.3b — Ultrasonic sensors at 75 cm (Known: 75.0 cm)

5.5 Analysis of Results

5.5.1 ToF Sensor Behaviour

Across all materials and distances, the VL53L0X ToF sensor exhibited a consistent negative bias — it systematically under-reads the true distance. After calibration, the average error was approximately −1.8 cm. The following material-specific observations were made:

Material	Behaviour	Reason
Wood	Slight under-read (~1.5 cm)	Diffuse IR reflection — some scatter
Metal	Slight under-read (IR specular)	Angled specular reflection reduces return signal
Glass	Significant under-read (up to 4.7 cm at 75 cm)	IR partially transmits through transparent surface
Concrete/Wall	Most accurate (~0.1 cm error)	Ideal diffuse reflector for IR
Coloured Plastic	Varies with colour	Darker colours absorb more IR
Fabric	Inconsistent	IR scatters unpredictably on fibrous surface

Table 5.5.1 ToF Sensor Behaviour

5.5.2 Ultrasonic Sensor Behaviour

The fused Kalman output of the two ultrasonic sensors was consistently accurate across most materials, staying within ±0.5 cm of the known distance in most cases. The following observations were made:

Material	Behaviour	Reason
Wood	Accurate (±0.3 cm)	Good acoustic reflector

Material	Behaviour	Reason
Metal	Accurate (±0.2 cm)	Hard flat surface, clean echo return
Glass	Accurate (±0.2 cm)	Sound reflects off surface regardless of transparency
Concrete/Wall	Most accurate (±0.1 cm)	Large, flat, hard — ideal acoustic surface
Coloured Plastic	Accurate (±0.4 cm)	Colour does not affect acoustic reflection
Fabric	Over-reads by up to 1.2 cm	Fabric absorbs some sound energy, weakening echo

Table 5.5.2 Ultrasonic Sensor Behaviour

5.5.3 Bias Direction and Safety Implication

A critical observation from the results is the consistent directionality of each sensor's error:

Sensor	Bias Direction	Average Magnitude	Safety Implication
ToF (VL53L0X)	Negative (under-reads)	~1.8 cm	Safe — drone reacts earlier than necessary
Ultrasonic (fused Kalman)	Positive (over-reads)	~0.6 cm	Risky — drone may not react in time

Table 5.5.3 Observations

For reactive obstacle avoidance, overestimation is the critical failure mode: if the system believes an obstacle is farther away than it truly is, the drone continues flying and a collision occurs. Underestimation is the safe failure direction — the drone slows or stops earlier than needed, which is conservative but does not result in damage. This bias analysis directly motivates the fusion weight assignment described in the following section.

5.6 Weighted Sensor Fusion

5.6.1 Weight Assignment

Fusion weights were assigned based on the bias direction and magnitude observed in the experimental results, with the objective of producing a fused output that is conservative — i.e., biased toward reporting obstacles as closer rather than farther.

$$\text{Fused Distance} = (0.65 \times \text{ToF_kalman}) + (0.35 \times \text{Ultrasonic_kalman})$$

The ToF sensor is assigned 65% weight because its consistent underestimation bias produces a conservative fused output — the fused distance will tend to be slightly shorter than the true distance, causing the drone to react earlier. The ultrasonic sensor retains a 35% weight as a cross-check, particularly for materials where the ToF sensor degrades (glass, fabric).

5.6.2 Safety Override Rule

The base weights assume both sensors are functioning reliably. When a sensor encounters a material outside its reliable operating range — such as the ToF sensor on glass, or the ultrasonic sensor on fabric — it may diverge significantly from the true distance. To handle this, a safety override rule was implemented:

$$\text{if } |\text{ToF_kalman} - \text{Ultrasonic_kalman}| > 3 \text{ cm:}$$

$$\text{Fused Distance} = \min(\text{ToF_kalman}, \text{Ultrasonic_kalman})$$

When the two sensor outputs disagree by more than 3 cm, the base weights are suspended and the shorter (more conservative) reading is used directly. This prevents a single-sensor failure from producing a dangerously overestimated fused output. The 3 cm threshold was derived from the experimental data: the maximum disagreement observed under normal operating conditions was approximately 2.6 cm (fabric at 50 cm), leaving a margin before the override activates.

5.7 Summary

The complete pipeline — hardware interfacing, temperature-compensated calibration, independent per-sensor Kalman filtering, dual-ultrasonic Kalman fusion, and weighted ToF-ultrasonic fusion with safety override — was implemented and validated on the ESP32 platform. The key findings are:

- The fused ultrasonic Kalman output achieves ±0.5 cm accuracy across most materials and distances after calibration.
- The VL53L0X ToF sensor exhibits a consistent underestimation bias of approximately 1.8 cm, which is the safe failure direction for obstacle avoidance.
- Glass is the most challenging material for the ToF sensor, with errors up to 4.7 cm at 75 cm, consistently triggering the safety override.
- Fabric is the most challenging material for the ultrasonic sensor, causing over-reads of up to 1.2 cm due to sound absorption.
- Concrete and wood are the most reliable surfaces for both sensors and serve as ideal calibration references.
- The 65/35 ToF-ultrasonic weight split produces a fused output with a conservative underestimation bias, suitable for safe reactive obstacle avoidance.
- The 3 cm override rule handles the two primary single-sensor failure modes — ToF on glass and ultrasonic on fabric — without requiring manual material identification.

CHAPTER 06:

HARDWARE & SOFTWARE DESCRIPTIONS

To build and verify the reactive obstacle avoidance system, specific components were chosen to balance processing efficiency, signal accuracy, and overall cost. This section lists the core hardware parts, software tools, and financial requirements needed to assemble the prototype.

6.1 Hardware Components:

6.1.1 ESP32

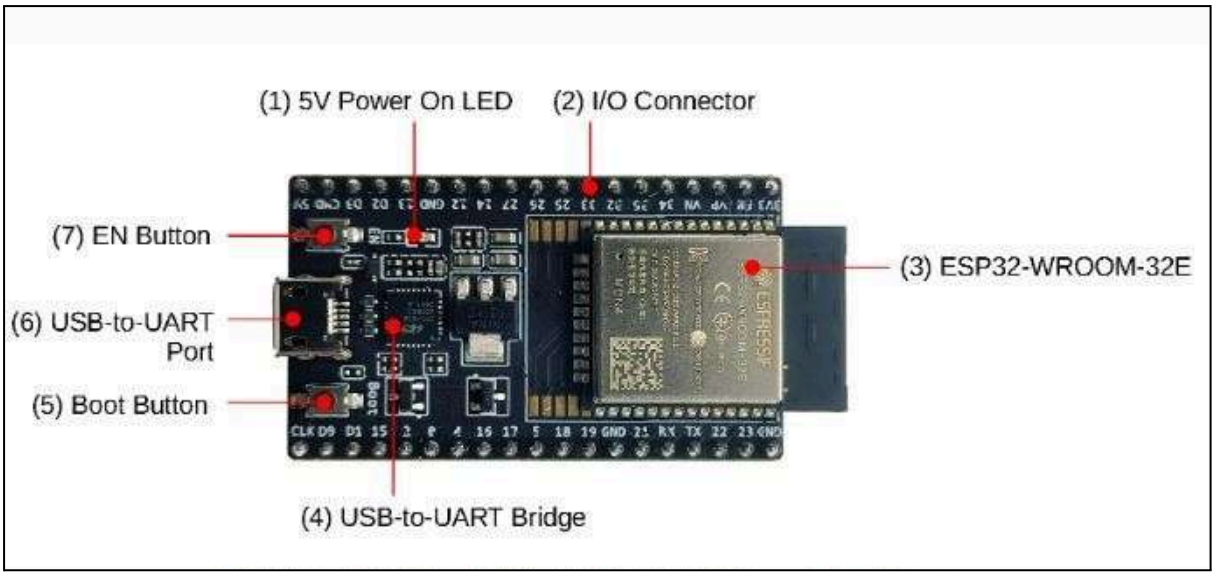


Fig 6.1.1 ESP32 Development Board

- ESP32 Development Board: The central computing unit of the platform, it features a 32-bit dual-core Xtensa LX6 microprocessor operating at clock speeds up to 240 MHz, providing the necessary computational speed to handle real-time execution loops without lag. Its native hardware supports dedicated I2C and multiple General Purpose Input/Output (GPIO) pins, allowing it to interface with the entire sensor grid simultaneously.

6.1.2 VL53L0X Time-of-Flight (ToF) Sensor:

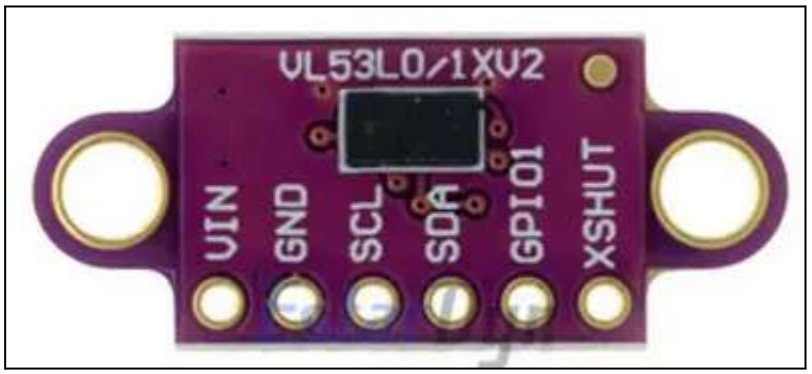


Fig 6.1.2 VL53L0X Time-of-Flight Sensor

- VL53L0X Time-of-Flight Sensor: A high-precision laser-ranging module that uses VCSEL (Vertical-Cavity Surface-Emitting Laser) technology to emit a 940 nm infrared pulse. It calculates distance by measuring the absolute travel time of light photons bouncing off an obstacle, providing localized spatial tracking independent of target color or reflectance. For obstacle avoidance, this sensor is heavily relied upon because its failure modes typically result in a safe underestimate.

6.1.3 HC-SR04 Ultrasonic Sensor:



Fig 6.1.3 HC-SR04 Ultrasonic Sensor

- HC-SR04 Ultrasonic Sensor: An acoustic transducer array consisting of an ultrasonic transmitter and receiver. It operates by transmitting a 40 kHz sonic pulse and measuring the width of the returning Echo pin pulse to determine distance based on

the speed of sound. This sensor serves as a vital cross-check against optically transparent surfaces like glass, where infrared light fails to reflect properly.

6.1.4 Resistors



Fig 6.1.4 Resistor

- Resistors (1 $k\Omega$,Quantity: 6) : Standard 0.25 W carbon film resistors configured as in-line voltage divider arrays. Because the HC-SR04 sensor operates on a 5V logic level while the ESP32 pins are strictly 3.3V tolerant, these resistors drop the high-voltage echo pulses down to a safe 3.33V level to prevent permanent hardware degradation.

6.1.5 Breadboard:

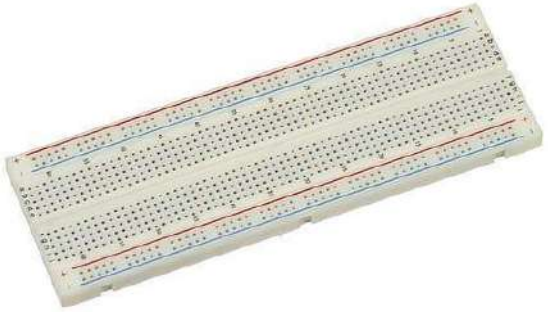


Fig 6.1.5 Breadboard

- A breadboard, solderless breadboard, or protoboard is a construction base used to build semi-permanent prototypes of electronic circuits. Unlike a perfboard or stripboard, breadboards do not require soldering or destruction of tracks and are hence reusable.
- Compared to more permanent circuit connection methods, modern breadboards have high parasitic capacitance, relatively high resistance, and less reliable

connections, which are subject to jostle and physical degradation.

- Solderless breadboards connect pin to pin by metal strips inside the breadboard. The layout of a typical solderless breadboard is made up from two types of areas, called strips. Strips consist of interconnected electrical terminals. Often breadboard strips or blocks of one brand have male and female dovetail notches so boards can be clipped together to form a large breadboard.

6.2 Software Components:

6.2.1 Arduino Integrated Development Environment (IDE):



Fig 6.2.1 Arduino Uno IDE

The development tools focus on direct firmware compilation and real-time data analysis:

- Arduino Integrated Development Environment (IDE): The foundational open-source toolchain used to write, compile, and upload the project firmware to the ESP32. It coordinates hardware timing registers to measure the pulse duration of the ultrasonic Echo pins, configures the Wire library to manage high-speed I2C communication with the VL53L0X, and compiles the real-time Kalman filtering math down to native machine code.

6.2.2 Data logging and dashboard:

- Data Logging Matrix : A systematic experimental spreadsheet application used to aggregate multi-sample data distributions during physical sensor profiling. It stores data per test distance (25 cm, 50 cm, and 75 cm) across all six distinct material targets, providing the structured historical datasets required to solve the least-squares linear regression calibration formulas.
- Interactive Sensor Fusion Dashboard : A front-end web application developed using HTML5, CSS3, and the Chart.js visualization engine. It ingests the Kalman-filtered and fused datasets to render dynamic absolute and signed error charts. The interface features responsive tab controls and zoom functions to visually verify under-reads, over-reads, and safety override boundaries across different materials in real time.

CHAPTER 07 :

PROJECT COST

Material	Quantity	Cost (INR)
ESP-32 Microcontroller	1	₹330
VL53L0X ToF Sensor	2	₹740
HC-SR04 Ultrasonic Sensor	2	₹300
Resistors (1 $k\Omega$)	6	₹60
Total Estimated Cost		₹1,430

Tab 7.1 Cost Estimation

IN WORDS: ONE THOUSAND FOUR HUNDRED AND THIRTY RUPEES ONLY.

CHAPTER 08 :

APPLICATION AND PROJECT STATUS

8.1 System Applications

By keeping the computational footprint minimal, this reactive obstacle avoidance system is highly adaptable for several real-world engineering environments:

By keeping the computational footprint minimal and processing data directly on the edge, this reactive obstacle avoidance system is highly adaptable for several real-world engineering environments:

- Low-Altitude Unmanned Aerial Vehicles (UAVs): Standard delivery, inspection, or surveillance drones operating at low altitudes face frequent hazards like power lines, tree branches, or building walls. This system provides an affordable, lightweight protective layer that handles safety boundaries without draining flight battery power on heavy computing.
- Indoor Navigating Robots: Automated Guided Vehicles (AGVs) and warehouse robots operate in dynamic spaces with frequent pedestrian traffic and changing layouts. The 3.0 cm safety override rule protects these units from colliding with problematic materials like glass partitions, clear plastic bins, or hanging fabric drapes.
- Low-Cost Industrial Automation: Instead of deploying expensive LiDAR or heavy camera arrays for simple point-to-point protection, this system offers a robust alternative using basic microcontroller architectures. It can be fitted to robotic arms or proximity gates to prevent accidents in shared human-machine workspaces.
- Close-Quarter Infrastructure Inspection: Drones inspecting tight spaces—such as the interiors of storage tanks, boilers, or under bridges—frequently lose GPS signals and struggle with dust or low lighting. Because this system relies entirely on local acoustic and optical sensors rather than external positioning or computer vision, it remains completely functional in dark, enclosed environments.
- Proximity Alert Networks for Heavy Machinery: Beyond fully autonomous platforms, this fusion logic can be integrated into operator-led industrial equipment, such as

- forklifts or mining vehicles. It acts as an active driver-assist system, giving reliable proximity warnings that filter out false positives from dust particles or exhaust steam.

8.2 Current Project Status

The engineering milestones for this project are structured across multiple phases, moving from initial validation to dynamic real-time deployment:

- Completed Testing and Analysis: We have successfully evaluated the VL53L0X Time-of-Flight and dual HC-SR04 ultrasonic sensors against six common environmental materials at the 25 cm, 50 cm, and 75 cm targets. Individual sensor biases and directional errors have been mapped.
- Software Implementation: The parallel digital Kalman filters, linear regression calibration code, base 65/35 weight distribution model, and the 3.0 cm safety override rule have been fully coded, debugged, and uploaded to the ESP32 development board.
- Data Visualization: The web-based sensor dashboard has been built, allowing us to interactively track absolute and signed errors across different surfaces to confirm the logic works seamlessly.

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